

Hospital Readmission Analysis

Insight Summary · Bharath Sattivelou · 2026

This short report sets out what I found when I looked into 30-day hospital readmissions using a dataset of just over a hundred thousand diabetic patient records. The aim was simple: work out which patients are most likely to come back to hospital within a month of being discharged, and what that might mean for the people trying to prevent it.

101,763

Patient records analysed

11.2%

Readmitted within 30 days

0.34

Top predictor coefficient

What the data showed

Out of every hundred patients in this dataset, roughly eleven ended up back in hospital within thirty days of leaving. That figure on its own is not surprising, but the interesting part was working out what set those patients apart from everyone else.

The single clearest signal was a patient's history. The number of times someone had already been admitted to hospital before their current stay turned out to matter far more than anything about that current stay itself. When I ranked all the factors by how strongly they pointed towards readmission, prior inpatient visits came out well ahead of the rest, with a weighting more than four times higher than the next factor down the list.

The headline finding: A patient's track record of previous admissions is a much better warning sign than how long their latest stay lasted or how many tests they had. In plain terms, readmission is less about the most recent visit and more about the wider pattern of someone's health over time.

The other factors that mattered

After prior admissions, two other things stood out, though to a lesser degree. The number of different diagnoses a patient had was the second strongest indicator, which makes sense when you think about it: people juggling several conditions at once are simply harder to keep stable once they go home. Being on diabetes medication also showed up as a contributing factor.

I expected the number of medications a patient was taking to be a bigger deal than it turned out to be. Patients who were readmitted were on slightly more medications on average than those who were not, but the gap was modest rather than dramatic. It is worth noting as a mild risk signal, but I would not want to lean on it too heavily on its own.

What I would do with this

If I were handing this to a hospital team, my recommendation would be straightforward. Rather than treating every discharge the same way, I would flag patients who already have a string of previous admissions behind them, and give that group extra attention before they leave: a proper medication review, a clear follow-up plan, and a check that support is in place at home. That is where the effort is most likely to pay off.

It is a more targeted approach than trying to do everything for everyone, and the data suggests it would catch the people most at risk without spreading resources too thin.

An honest note on the model

I built a logistic regression model to predict readmission, and it is worth being upfront about its limits. Because readmissions are relatively rare (only around one in nine patients), the model finds the majority group easy to predict but struggles more with the smaller group of patients who actually do come back. This is a common challenge with imbalanced data of this kind.

In a real production setting, the next step would be to either rebalance the data or move to a more sophisticated method such as gradient boosting, both of which would likely improve how well the model catches the at-risk minority. For the purpose of understanding the drivers of readmission, though, the analysis does its job and the findings hold up.